Seong Hyeon Kim

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Education

Pohang University of Science and Technology (POSTECH), Ph.D. in Electrical Engineering — Feb 2026 – Present
- Advisor: Prof. Hyemin Ahn | Lab: Robot Learning Lab (RLLab)

Ulsan National Institute of Science and Technology (UNIST), M.S. in Artificial Intelligence — Mar 2024 – Feb 2026
- Advisor: Prof. Hyemin Ahn | Lab: AI-HRI Lab

Myongji University, B.S. in Mechanical Engineering — Mar 2018 – Feb 2024

Experience

SuperbAI, Research Intern — Dec 2025 – Feb 2026
- Project: **Text-to-Humanoid Control**
- Developed motion synthesis algorithms for humanoid robots conditioned on natural language.

Research Experience

Graduate Student Researcher, Robot Learning Lab (RLLab) – POSTECH — Feb 2026 – Present

Graduate Student Researcher, AI-HRI Lab – UNIST — Mar 2024 – Feb 2026
- Focusing on **Robotic Manipulation** and **World Models** for intelligent robot control.

Undergraduate Intern, Dynamic Robot Control Lab (DRCL) – Myongji University — Mar 2019 – Feb 2024
- Conducted research on mechanical design and parallel mechanisms for wheel-legged robots.

Publications

Proprioception-conditioned Visual Scene Generation for Robot World Modeling via Contrastive Learning and Diffusion — 2025
S. H. Kim, H. Ahn | *Journal of ICROS*

Design of a Wheel-Legged Robot using Linear 2-DOF Parallel Mechanism — 2023
S. H. Kim, D. Choi | *KSME Conference*

Application of TDOA Algorithm for Sound Source Direction Tracking — 2023
T. Kim, **S. H. Kim**, D. Choi | *KSME Conference*

2-DOF Parallel Mechanism for Linear Wheel-Legged Robot System — 2023
S. H. Kim, T. Kim, S. Ryu, J. Han, D. Choi | *KSME Conference*

Ball Tracking and Trajectory Measurement System for Tennis Robot — 2022
Y. Yang, S. Ahn, **S. H. Kim**, D. Choi | *KSME Conference*

Real-time Object Trajectory Prediction using AI and Stereo Vision — 2022
Y. Yang, S. Ahn, **S. H. Kim**, D. Choi | *KSME Conference*

Honors & Awards

Bronze Award, School Capstone Design Project – Myongji Univ. (Elderly Companion Robot) — Dec 2023
Bronze Award, Hanium ICT Mentoring – KFII (Elderly Companion Robot) — Dec 2023
Silver Award, SEP Convergence Festival – Myongji Univ. (Elderly Companion Robot) — Dec 2023
Encouragement Award, Capstone Design Competition – KSME (Sound Source Tracking) — Apr 2023

Best Paper Award, Spring Conference – KSME (Tennis Ball Tracking) Apr 2023
Bronze Award, Capstone Design Competition – Myongji Univ. (Desert Seeding Robot) Dec 2019

RLLAB team, 4th place - 2026 Humanoid Challenge Jul 2026

Professional Service

Reviewer, IEEE International Conference on Robotics and Automation (ICRA) 2025
Volunteer, LG Project-based Learning Program Summer 2025

Technical Skills

Programming: Python, C, MATLAB

Tools: ROS/ROS2, MuJoCo, Gazebo, PyTorch, Git, SolidWorks, KiCad, LaTeX

Robots: Unitree G1, Unitree Go2, Universal Robots (UR3, UR5), Franka Emika Panda

Technical Report

Demonstrating a Vision-Based AI Robot for Strategic Board Games

Taehwan Kim*, Dokeun Lee*, Seonghyeon Kim*, Yongjun Choi*, Sungjun Heo, Thi Thuy Ngan Duong, Kyungdon Joo, Namhun Kim, Jeong hwan Jeon, Hyemin Ahn

* The first four authors contributed equally.

Low-cost Gomoku robot with real-time vision, reinforcement learning decisions, and robotic arm control.

Project website